

T-030 frontier consolidation – arbitrary-Q DR-2: statement map, an

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

T-030 is the arbitrary- Q (non-lattice) DR-2 problem: for every finite antipodal $Q \subset S^2$ with $|Q| = N$, is the additive energy

$$E_+(Q) := \#\{(a, b, c, d) \in Q^4 : a + b = c + d\}$$

bounded by $N^{2+\epsilon}$ for every $\epsilon > 0$? This is the content of B5's hypothesis H-NONLATTICE-REMAINDER-EXCLUDED – the last remaining B5 T7 blocker after the 2026-06-12 operator verdicts (D1-A/D2-A/D3-A). Boundary kept explicit throughout (main-line-synthesis v1.3 Sec. 4b): this track is NOT load-bearing for the published C_{full} head theorem, whose Lemma 2 caps $T' \leq \lfloor 2\pi/\theta_{\min} \rfloor = 10$ elementarily in-class; T-030 is a frontier strengthening. This note consolidates the frontier (Sec. 2), proves one new unconditional-modulo-classical intermediate exponent (Sec. 3), discharges the R1 residual of the conditional chain by a written proof (Sec. 4), and records the pre-registered machine experiments including one honestly-fired gate (Sec. 5). No gate or tier action is taken.

2. Statement map of the frontier (consolidation)

2.1 The exact open statement

Open (T-030). For every finite antipodal $Q \subset S^2$, $|Q| = N$: $E_+(Q) \lesssim_\epsilon N^{2+\epsilon}$. (Antipodality plays no role in any partial result below; everything stated holds for arbitrary finite Q .)

2.2 What is proved, at what grade (pre-this-note)

- **Lemma A (R-025), T7 unconditional:** $E_+(Q) \leq (1 + T'(Q))N^2$, where $T'(Q)$ is the maximum occupancy of a proper sum-level circle $C_m = S^2 \cap \{x \cdot m = \frac{1}{2}|m|^2\}$, $m \neq 0$. Hence $T' = O(1)$ classes (random, great circle) satisfy DR-2 elementarily.
- **Lattice class (R-026 + Lemma NT), T7-NTstandard:** for $Q \subset \mathbb{Z}^3 \cap \{|x|^2 = R\}$, $T' \lesssim_\epsilon R^\epsilon$ (divisor + Dirichlet class number), so $E_+ \leq (1 + C_\epsilon R^\epsilon)N^2$. SOLVED; this is what discharged H-ADM-COH for the lattice class.
- **Separated Q , T6 PROVED CONDITIONAL on Bourgain–Demeter ($d = 3, p = 4$):** via the Besicovitch mean $M(|f_Q|^4) = E_+(Q)$ and the Schwartz-majorant bridge (dr2-decoupling-closure v1.1).
- **Arbitrary Q , conditional chain, T4+/T5-candidate:** the cross-scale induction (dr2-cross-scale-induction v1.1) reduces arbitrary finite Q to the separated base via affine rescaling + decoupling stability, with TWO cited-not-reproduced residuals: R1 (local-to-global translate-averaging) and R2 (multi-scale ϵ -bookkeeping).

- **PSM route:** Pencil Rigidity / Pair-Sum Surface Multiplicity conjecture T2 (falsification gate $E_+(Q_N) \geq N^{2+\delta}$); reduction $\text{PSM} \Rightarrow \text{DR-2}$ at T3 with OPEN GAPS α (decomposition existence) and β (per-cluster uniformity). Cross-energy lemma R-021 (T7): $E_{\text{off}}(A, B) \leq 2 \min(|A|^2 - |A|, |B|^2 - |B|)$ for circle subsets.
- **Affine invariance (R-023), T7:** E_+ is exactly affine-invariant; cap clustering cannot beat N^2 by rescaling alone.
- **Unconditional record for arbitrary Q (pre-this-note):** the trivial/triple-count $N^{5/2}$ class. Lemma A gives only $(1 + T')N^2 \leq N^3$ in general.

2.3 Why the rich latitude blocks the naive T' route

The engineered latitude circle (N points on $z = z_0$) has ONE maximally rich sum-circle: $m^* = (0, 0, 2z_0)$ with C_{m^*} equal to the latitude itself, so $T' = N$ – yet $E_+ \approx 3N^2$ (machine-exact: $2.972N^2$ at $N = 108$, Sec. 5). So $T' \lesssim_\epsilon N^\epsilon$ is SUFFICIENT for DR-2 (Cor. 2 of R-025) but NOT necessary, and no bound on T' alone can close T-030. The missing ingredient is visible in the witness: what matters is not how rich the richest circle is, but HOW MANY simultaneously rich circles can coexist. That is an incidence question, and Sec. 3 exploits exactly this.

2.4 What an unconditional proof must overcome

By the structure corollary of Sec. 3 (Cor. 1.2), any family with $E_+ \geq N^{2+\delta}$ must contain $\gtrsim N^{2+\delta}/(k^2 \log N)$ simultaneously k -rich sum-circles at some single dyadic richness $k \in [c_1 N^\delta, c_2 N^{(1-\delta)/3}]$. Classical incidence bounds cap what this regime can do at $\delta = 1/4$ (Pach–Sharir) or $2/9 + \epsilon$ (Aronov–Sharir); pushing $\delta \rightarrow 0$ unconditionally would need near-optimal circle-incidence technology (the regime adjacent to the hard unit-distance-type problems), while the conditional Bourgain–Demeter route bypasses incidences entirely at the price of the R2 bookkeeping.

3. Theorem 1: the unconditional $N^{9/4}$ bound (new)

3.1 The incidence input

[CIRC-INC] (classical; Clarkson–Edelsbrunner–Guibas–Sharir–Welzl 1990; Pach–Sharir curves with 3 degrees of freedom). There is an absolute constant A such that for any N points and any n DISTINCT circles in the plane, the number of point-circle incidences satisfies

$$I \leq A(N^{3/5}n^{4/5} + N + n).$$

Transfer to S^2 : stereographic projection from any $p \in S^2$ avoiding the finitely many circles and points (such p exists; finitely many circles have measure zero) maps the configuration bijectively to points and distinct circles in the plane, preserving incidences. So [CIRC-INC] holds verbatim for circles on the sphere.

3.2 Setup and the dyadic decomposition

As in R-025: $r(m) = \#\{(a, b) \in Q^2 : a + b = m\}$, $E_+(Q) = \sum_m r(m)^2$, $\sum_m r(m) = N^2$. For $m \neq 0$ both summands of $a + b = m$ lie on the sum-level circle C_m , $r(m) \leq t_m := \#(Q \cap C_m)$, and the map $m \mapsto C_m$ is INJECTIVE on $\{0 < |m| \leq 2\}$: the plane $x \cdot m = \frac{1}{2}|m|^2$ has unit normal $m/|m|$ and distance $|m|/2 > 0$ from the origin; a circle on S^2 determines its plane, and the pair (normal, distance > 0) determines $m = 2du$. Distinct m therefore give distinct circles, as [CIRC-INC] requires.

Let $n_k := \#\{m \neq 0 : r(m) \geq 1, t_m \geq k\}$ (a finite family of distinct k -rich circles). Dyadic classes: $D_j = \{m \neq 0 : r(m) \geq 1, 2^j \leq t_m < 2^{j+1}\}$, $j \geq 0$.

Per-class bounds. For $k = 2^j$, $j \geq 1$:

$$(i) \quad \sum_{m \in D_j} r(m)^2 \leq 2k \sum_{m \in D_j} r(m) \leq 2kN^2, \quad (ii) \quad \sum_{m \in D_j} r(m)^2 \leq 4k^2 |D_j| \leq 4k^2 n_k.$$

The $j = 0$ class has $r(m) = t_m = 1$, contributing $\leq N^2$; the $m = 0$ term contributes $r(0)^2 \leq N^2$.

Richness count from [CIRC-INC]. Each of the n_k circles carries $\geq k$ incidences, so $k n_k \leq I \leq A(N^{3/5} n_k^{4/5} + N + n_k)$. For $k \geq 2A$ the An_k term is absorbed ($An_k \leq \frac{k}{2} n_k$) and one of the two remaining terms dominates, giving

$$n_k \leq (4A)^5 \frac{N^3}{k^5} + 4A \frac{N}{k} \quad (k \geq 2A).$$

3.3 Theorem 1 and proof

Theorem 1. There is an absolute constant C (explicit in the [CIRC-INC] constant A) such that for EVERY finite $Q \subset S^2$,

$$E_+(Q) \leq C N^{9/4}.$$

Proof. Split the dyadic sum at $k^* = N^{1/4}$:

$$E_+(Q) \leq 2N^2 + \sum_{1 \leq 2^j \leq k^*} \sum_{m \in D_j} r(m)^2 + \sum_{2^j > k^*} \sum_{m \in D_j} r(m)^2.$$

Small classes by (i): $\sum_{2^j \leq k^*} 2 \cdot 2^j N^2 \leq 4k^* N^2 = 4N^{9/4}$. Large classes by (ii) and the richness count (the finitely many dyadic k with $k^* < k < 2A$, possible only when $N^{1/4} < 2A$, contribute $\leq 2(2A) \cdot 2AN^2 = O_A(N^2)$ by (i)):

$$\sum_{2^j > k^*} 4k^2 n_k \leq \sum_{2^j > k^*} \left(4(4A)^5 \frac{N^3}{k^3} + 16A k N \right) \leq \frac{32}{7} (4A)^5 \frac{N^3}{(k^*)^3} + 32A T' N \leq C' N^{9/4} + 32A N^2,$$

using $\sum_{k > k^*} \text{dyadic } k^{-3} \leq \frac{8}{7} (k^*)^{-3}$, $T' \leq N$, and $N^3 / (k^*)^3 = N^{9/4}$. Summing, $E_+(Q) \leq C N^{9/4}$. $\square$

Remark 1.1 (Aronov–Sharir refinement). Replacing [CIRC-INC] by the Aronov–Sharir circle bound $I \lesssim N^{2/3} n^{2/3} + N^{6/11} n^{9/11} \log^{2/11}(N^3/n) + N + n$ gives $n_k \lesssim N^2/k^3 + N^3 k^{-11/2} \log^{O(1)} + N/k$ and, balancing at $k^* = N^{2/9}$, the exponent $E_+(Q) \lesssim_\epsilon N^{20/9+\epsilon}$. The 9/4 statement is kept as the headline because its input is the cleaner classical bound; both are strictly below the prior $N^{5/2}$ record.

Corollary 1.2 (structure of potential extremizers). If $E_+(Q) \geq N^{2+\delta}$ for some $\delta > 0$ and N large, then some single dyadic class contributes $\geq N^{2+\delta} / (2 \log_2 N)$, and by (i), (ii) + [CIRC-INC] its richness scale k obeys

$$c_1 \frac{N^\delta}{\log N} \leq k \leq c_2 N^{(1-\delta)/3} \log^{1/3} N, \quad n_k \geq \frac{N^{2+\delta}}{8k^2 \log_2 N},$$

forcing $\delta < 1/4$ (consistency of the window) and a SUPERLINEAR-in- N number $n_k \geq N^{2+\delta} / (8k^2 \log N) \geq N^{4/3+O(\delta)}$ of simultaneously k -rich sum-circles. The rich latitude (ONE rich circle) and all tested families are nowhere near this regime (Sec. 5).

Honest grade. Theorem 1 is PROVED MODULO [CIRC-INC]: every step above is self-contained EXCEPT the incidence input, which is classical published combinatorial geometry (CEGSW 1990 / Pach–Sharir), invoked the way R-026 invokes textbook number theory (T7-NTstandard precedent). Whether the repository registers this as the same modulo-classical grade is an OPERATOR decision; no enactment here. Within the repo this improves the unconditional arbitrary- Q record from the $N^{5/2}$ class to $N^{9/4}$ ($\delta = 1/4$); the statement itself is assembled from standard tools and no novelty beyond the repository record is claimed.

4. Lemma R1: the local-to-global bridge, now a written proof

dr2-cross-scale-induction v1.1 rested on two CITED residuals; R1 was the decoupling-compatible passage from the ball inequality to the Besicovitch-mean energy inequality. R1 is in fact provable outright for finite Q , because $|f_Q|^4$ is a trigonometric polynomial. We record the proof.

Lemma R1 (translate-averaged decoupling $\Rightarrow$ energy inequality). Let $Q \subset S^2$ be finite, $\{\theta\}$ a finite partition of S^2 into caps, $Q_\theta = Q \cap \theta$, and $f_P(x) = \sum_{q \in P} e(q \cdot x)$. Suppose the LOCAL decoupling inequality holds: for some ball radius R_0 , weight $0 \leq w \in \mathcal{S}(\mathbb{R}^3)$ with $\int w > 0$ and $\hat{w}$ absolutely summable over the (finite) relevant frequency set, and constant D ,

$$\|f_Q\|_{L^4(w(\cdot-y))} \leq D \left(\sum_{\theta} \|f_{Q_\theta}\|_{L^4(w(\cdot-y))}^2 \right)^{1/2} \quad \text{for every translate } y \in \mathbb{R}^3. \quad (4.1)$$

Then

$$E_+(Q) \leq D^4 \left(\sum_{\theta} E_+(Q_\theta)^{1/2} \right)^2.$$

Step 0 (one inequality suffices for all translates). If (4.1) is assumed only at $y = 0$ (the usual statement on B_{R_0}), it holds for every y with the SAME D : substituting $g_P(x) = f_P(x + y) = \sum_q e(q \cdot y) e(q \cdot x)$ preserves the frequency support and cap decomposition (unimodular coefficients; decoupling constants are insensitive to them), and $\|f_P\|_{L^4(w(\cdot-y))} = \|g_P\|_{L^4(w)}$.

Step 1 (exact translate-averaging identity). $|f_Q(x)|^4 = \sum_v c_v e(v \cdot x)$ is a FINITE sum with $c_v = \#\{(a, b, c, d) \in Q^4 : a + b - c - d = v\} \geq 0$ and $c_0 = E_+(Q)$. For $S > 0$, by Fubini (all sums finite),

$$\text{avg}_S := \frac{1}{|B_S|} \int_{B_S} \|f_Q\|_{L^4(w(\cdot-y))}^4 dy = \sum_v c_v \hat{w}(v) \beta_S(v), \quad \beta_S(v) := \frac{1}{|B_S|} \int_{B_S} e(v \cdot y) dy,$$

with $\beta_S(0) = 1$ and $|\beta_S(v)| \leq 3/(2\pi|v|S)^2 \cdot (1+o(1)) \rightarrow 0$ for $v \neq 0$ (explicitly $\beta_S(v) = 3(\sin u - u \cos u)/u^3$, $u = 2\pi|v|S$). Hence

$$\lim_{S \rightarrow \infty} \text{avg}_S = \hat{w}(0) c_0 = \left(\int w \right) E_+(Q),$$

and identically, for each cap, $\lim_S \frac{1}{|B_S|} \int_{B_S} \|f_{Q_\theta}\|_{L^4(w(\cdot-y))}^4 dy = (\int w) E_+(Q_\theta)$. (Machine check: dr2_t030_r1_bridge.py verifies $c_0 = E_+$ as two independent exact integer counts and the convergence with the predicted S^{-2} envelope, final relative error $\leq 2.4 \times 10^{-5}$ at $S = 80$ on both test configurations.)

Step 2 (Minkowski). Set $A_\theta(y) = \|f_{Q_\theta}\|_{L^4(w(\cdot-y))}^2$. Raising (4.1) to the 4th power and averaging over $y \in B_S$,

$$\text{avg}_S \leq D^4 \left\| \sum_{\theta} A_\theta \right\|_{L^2(\mu_S)}^2 \leq D^4 \left(\sum_{\theta} \|A_\theta\|_{L^2(\mu_S)} \right)^2, \quad \mu_S = \frac{dy|_{B_S}}{|B_S|},$$

by the triangle inequality in $L^2(\mu_S)$ applied to the FINITE sum over caps. And $\|A_\theta\|_{L^2(\mu_S)}^2 = \frac{1}{|B_S|} \int_{B_S} \|f_{Q_\theta}\|_{L^4(w(\cdot-y))}^4 dy$.

Step 3 (limit). Let $S \rightarrow \infty$ in Step 2: the left side tends to $(\int w) E_+(Q)$ by Step 1; on the right, each of the finitely many cap terms tends to $[(\int w) E_+(Q_\theta)]^{1/2}$. Dividing by $\int w > 0$ gives the boxed inequality. $\square$

Consequence for the conditional chain. In dr2-cross-scale-induction v1.1, the iteration inequality $E_+(Q) \lesssim_{\epsilon} \delta^{-\epsilon} (\sum_{\theta} E_+(Q_\theta)^{1/2})^2$ now follows from the Bourgain–Demeter ball inequality alone via Lemma R1 – the local-to-global passage is NO LONGER a cited residual. The chain for arbitrary Q is now conditional on exactly: (a) the Bourgain–Demeter ball decoupling black box (with its C^2 -stability for the rescaled caps, as already recorded), and (b) residual R2 (multi-scale ϵ -bookkeeping over $O(\log N)$ levels). Whether {black box + R2} warrants moving the chain from T4+/T5-candidate is an OPERATOR decision; the factual change here is the residual set shrinking $\{R1, R2\} \rightarrow \{R2\}$.

PSM gaps (status note). The PSM route’s gaps α (decomposition existence) and β (per-cluster uniformity) are untouched by this note; the adversarial scan of Sec. 5 doubles as the PSM falsification gate scan and found NO candidate family (all scaling exponents ≈ 2). Theorem 1 supersedes the PSM route as the best unconditional statement: PSM would give $N^2 \log^B N$ but is T2, while $N^{9/4}$ is proved modulo [CIRC-INC] only.

5. Machine experiments (pre-registered gates; one honest firing)

Scripts: codes/vacuum/dr2_t030_dyadic_richness.py (v1.1.0, 9/9 asserts) and codes/vacuum/dr2_t030_r1_bridge.py (v1.0.0, 3/3 asserts); exact integer arithmetic throughout (every finite-precision configuration is a scaled-rational one, so exact experiments are necessarily of this form; the theorems cover arbitrary real Q). Gates G1–G6 and H1–H3 were declared in the scripts BEFORE the first full run and are quoted in the JSON artefacts.

Families (adversarial; the known $T' = N$ witness as control): F1 rich latitude (108 points of an r_2 -rich Gaussian-integer circle, $a^2 + b^2 = 1105^2$, at $z_0 = 47$); F1b its antipodal double ($N = 216$); F2 random antipodal subset of the Gauss-typical shell $R = 9974$ ($N = 150$); F3 antipodal two-cap cluster ($N = 120$); F4 union of the 4 richest latitudes of the engineered sphere ($N = 316$; Cayley-type: each circle is a unit/prime angle orbit); F5 great circle ($N = 108$); F6 full shell $R = 4994$ ($N = 1152$).

Results (all exact):

family	N	E_+	E_+/N^2	T'	$E_+/N^{9/4}$
F1 rich latitude	108	34668	2.972	108	0.922
F1b antipodal double	216	208008	4.458	108	1.163
F2 random antipodal	150	67194	2.986	7	0.853
F3 two-cap cluster	120	48600	3.375	16	1.020
F4 four rich latitudes	316	227900	2.282	108	0.541
F5 great circle	108	34668	2.972	2	0.922
F6 full shell $R=4994$	1152	5209344	3.925	32	0.674

G1 estimator audit (vectorized $E_+ =$ brute-force $O(N^4)$ count on three small instances) PASS; G2 Lemma A PASS on all families; G3 ($r(m) \leq t_m$, the Theorem-1 proof step) PASS on all proper sums of all families; G4 (Theorem-1 consistency, $E_+ \leq 5N^{9/4}$) PASS, max ratio 1.163; G6 (per-class dyadic bookkeeping inequalities) PASS on every class of every family. H1–H3 (Lemma R1 ingredients) PASS as quoted in Sec. 4.

Honest gate-firing record (G5). The pre-registered PSM proxy (any family with $N \geq 100$ and $E_+ \geq N^{2.2}$ must be reported) FIRED on five families, worst finite- N exponent 2.278 (F1b). Investigation per the registered protocol (G5b, added v1.1.0 BEFORE resolution): rebuilding the same antipodal-double-latitude construction at $3\times$ size (r_2 -richer circle $a^2 + b^2 = 32045^2$, $N = 648$) gives E_+/N^2 $4.458 \rightarrow 4.486$ and scaling exponent $\log(E_2/E_1)/\log(N_2/N_1) = 2.0057$. Verdict: the firing is a CONSTANT-FACTOR artifact of the finite- N proxy (a constant $c \approx 4.5$ at $N = 216$ mimics exponent 2.28); growth is quadratic; NO PSM/DR-2 counterexample candidate. Lesson recorded: finite- N exponent proxies must be paired with scaling tests at registration time.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

- (α) *Theorem 1’s [CIRC-INC] is cited, not reproduced – the same defect class as citing decoupling; why grade it differently?* VALID with mitigation: it IS a citation, and the note says so in the grade line itself (PROVED MODULO [CIRC-INC]). The mitigation is the precedent class: CEGSW/Pach–Sharir is classical 1990s combinatorial geometry of the same standing as the textbook number

theory behind R-026's T7-NTstandard, whereas Bourgain–Demeter is modern heavy machinery whose internal bookkeeping (R2) is precisely what the conditional chain cannot reproduce. The operator decides whether the repository accepts the analogy; nothing is enacted here.

- (β) *The incidence bound needs DISTINCT circles; different m might give the same circle C_m , collapsing n_k .* DISMISSED with explicit counter: the plane of C_m has unit normal $m/|m|$ at distance $|m|/2 > 0$; a circle determines its plane, and (normal, positive distance) determines $m = 2du$ – the map $m \mapsto C_m$ is injective on $0 < |m| \leq 2$ (Sec. 3.2). The only degenerate case $m = 0$ is split off separately, exactly as in Lemma A.
- (γ) *Lemma R1 Step 0 assumes the decoupling constant is uniform over translates.* DISMISSED: translation $x \mapsto x + y$ multiplies each frequency atom by the unimodular constant $e(q \cdot y)$, preserving frequency supports and all cap norms; (4.1) at $y = 0$ therefore transfers to every translate with the same D . No new hypothesis enters.
- (δ) *Step 2/3 order of limits: Minkowski at finite S , then $S \rightarrow \infty$ – any illegal interchange?* DISMISSED: at each finite S the inequality involves a FINITE sum over caps in L^2 of a probability measure; the $S \rightarrow \infty$ limit is then taken of both sides, each cap term converging by the Step-1 identity (finite Fourier sums; no dominated convergence or infinite-sum interchange is invoked anywhere).
- (ϵ) *Gate G5 fired – does the no-counterexample claim survive?* VALID with mitigation, fully recorded in Sec. 5: the registered proxy fired, the registered consequence (report + investigate) was executed, and the scaling test (exponent 2.0057) resolves it as a constant-factor artifact. The claim made is exactly what the data supports: no tested family shows super-quadratic GROWTH.

Quantitative sanity check (mandatory). Magnitude: at $N = 216$, $N^{9/4} \approx 1.79 \times 10^5$ vs $E_+ = 208008$, ratio 1.163 – the Theorem-1 constant C must exceed 1.16, consistent with the proof's $C = O(A^5)$ (A the [CIRC-INC] constant; no contradiction). Limit case: great circle saturates Lemma A, $E_+/N^2 = 2.972$ vs ceiling $(1 + T') = 3$ – the $m = 0$ split is necessary and the estimator is calibrated. Exponent direction: $E_+/N^{9/4}$ DECREASES along growing controls (F6 full shell 0.674 at $N = 1152$; F1b scaling point E_+/N^2 flat at 4.49), as an N^2 -class truth under an $N^{9/4}$ bound requires. Convergence rate: Lemma R1 machine check error $1.8\text{--}2.3 \times 10^{-5}$ at $S = 80$ sits below the $3/u_{\min}^2 = 1.07 \times 10^{-4}$ envelope with $u_{\min} = 2\pi S/3$ – the predicted S^{-2} mechanism, not accidental cancellation. Reproducibility: both scripts are deterministic (fixed seeds), exact-integer in every energy count, and exit 0 only if all 12 asserts pass.

7. Result footer

Result ID:	R-030 (Theorem 1: unconditional-modulo-[CIRC-INC] $N^{9/4}$ bound) + R-031 (Lemma R1: local-to-global bridge written proof) / T-030 frontier consolidation
Precise statement:	(1) For every finite Q in S^2 , $E_+(Q) \leq C N^{9/4}$, proved via Lemma A's sum-circle structure + dyadic occupancy classes + the classical circle-incidence bound [CIRC-INC] (CEGSW/Pach-Sharir, $s=3$); with Aronov-Sharir, $N^{20/9+\epsilon}$. Corollary: any $E_+ \geq N^{2+\delta}$ family needs $\geq N^{2+\delta}/(8k^2 \log N)$ simultaneously k -rich sum-circles at some k in $[N^\delta/\log, N^{(1-\delta)/3} \log^{1/3}]$, forcing $\delta < 1/4$. (2) Lemma R1: the ball decoupling inequality alone implies $E_+(Q) \leq D^4 (\sum_{\theta} E_+(Q_{\theta})^{1/2})^2$ (exact Besicovitch limit for trig polynomials + Minkowski in $L^2(dy)$); the cross-scale chain's cited residual set shrinks

$\{R1, R2\} \rightarrow \{R2\}$. (3) Statement map + adversarial scan: no super-quadratic-growth family found (G5 proxy fired, resolved as constant-factor artifact, scaling exponent 2.0057).

Scope: T-030 (arbitrary non-lattice finite Q in S^2 ; B5 hypothesis H-NONLATTICE-REMAINDER-EXCLUDED). NOT load-bearing for the published C_{full} head (Lemma 2 caps $T' \leq 10$ in-class; synthesis v1.3 Sec. 4b). Antipodality not needed for any partial result.

Evidence grade: Theorem 1: PROVED MODULO [CIRC-INC] (classical published incidence geometry; operator may grade analogously to R-026's T7-NTstandard). Lemma R1: written unconditional proof GIVEN the ball decoupling inequality (the named black box); its own two ingredients are self-contained. Conditional chain: T4+/T5-candidate with residual set now $\{R2\}$ -- any regrade is the operator's. Open statement $E_+ \leq_{\epsilon} N^{2+\epsilon}$: STILL OPEN.

Reproduction command: `python3 codes/vacuum/dr2_t030_dyadic_richness.py`
`python3 codes/vacuum/dr2_t030_r1_bridge.py`

Expected output: 9/9 and 3/3 asserts PASS, exit 0; JSON artefacts at `claims/B5-BEYOND-LAYER-BOUND/runs/260612-dr2-t030-dyadic-richness/` and `.../260612-dr2-t030-r1-bridge/`.

Falsifier: (a) any finite Q with $E_+(Q) > C N^{9/4}$ beyond the proof constant (would refute Theorem 1 / [CIRC-INC]); (b) a family with SCALING exponent > 2 along its own construction (PSM/DR-2 counterexample; gate G5/G5b protocol); (c) failure of the S^{-2} Besicovitch convergence or $c_0 \neq E_+$ in the R1 bridge (gates H1/H2).

Tier before / after: NONE -- operator decides. No gate flip, no tier enactment, no hypothesis removal. B5 stays T6 PROVED-CONDITIONAL on $H_{B5}T6$; DR2-SHARE stays RESCOPED-T0-T030-NONLOADBEARING; T-030 stays OPEN.

No-overclaim statement: This note does NOT close T-030: $E_+ \leq_{\epsilon} N^{2+\epsilon}$ for arbitrary (antipodal) Q remains open. Theorem 1 is an INTERMEDIATE exponent (9/4), conditional on a cited classical incidence bound; no novelty beyond the repository record is claimed for it. Lemma R1 removes ONE of the two cited residuals of the conditional chain; the chain remains conditional on Bourgain-Demeter + R2. The G5 gate firing is recorded, not suppressed.

Next required action: Operator review: (i) whether Theorem 1's modulo-classical grade is registered (T7-INCstandard analogue) as the new unconditional T-030 record; (ii) whether the conditional chain's residual shrinkage $\{R1, R2\} \rightarrow \{R2\}$ updates its T4+/T5-candidate grade; (iii) research continuation: either reproduce R2 (multi-scale bookkeeping) for a T6 conditional chain closure, or push the incidence route below 20/9 (open incidence technology).